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higher boiling point compared to branch chain isomers. The boiling point decreases with increase in branching as the molecule becomes compact and the area of the contact decreases.

2) Solubility and density

Water molecules are polar and alkanes are non-polar. The insolubility of alkanes in water makes them good water repellent for metals which protects the metal surface from corrosion. Because of their lower density than water, they form two layers and occupy top layer. The density difference between alkanes and water explains why oil spills in aqueous environment spread so quickly.



Apples produce its own wax which contains long unbranched alkanes ($C_{27}H_{56}$ and $C_{29}H_{60}$). This natural wax helps the apple to resist moisture lost, enhances the fruit firmness and slowdowns its natural degradation. The consumption of apple with artificial wax coating is harmful to the digestive system



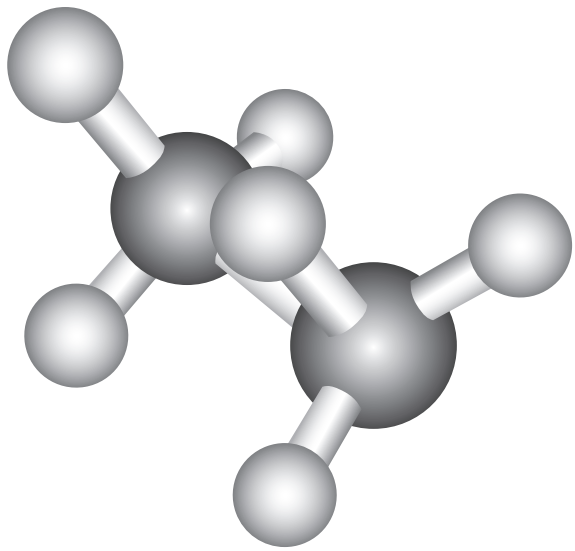
13.2.3 Conformations of alkane:

Each carbon in alkanes is sp^3 hybridized and the four groups or atoms around the carbon are tetrahedrally bonded. In alkanes having two or more carbons, there exists free rotation about C-C single bond. Such rotation leaves all the groups or atoms bonded to each carbon into an infinite number of readily interconvertible three dimensional arrangements. Such readily interconvertible three dimensional arrangement of a molecule is called conformations.

(i) Conformations of ethane:

The two tetrahedral methyl groups can rotate about the carbon – carbon bond axis yielding several arrangements called conformers. The extreme conformations are staggered and eclipsed conformation. There can be number of other arrangements between staggered and eclipsed forms and their arrangements are known as skew forms.

Eclipsed conformation:



In this conformation, the hydrogen's of one carbon are directly behind those of the other. The repulsion between the atoms is maximum and it is the least stable conformer.